EUROSIM 2026 Presentation Script

Slide 1 Introduction

Hello everyone. My name is Yu Tao, and today I will present our work on a task-oriented executable modeling framework for smart chemical laboratories.

This work examines how a laboratory task moves from a proposed intent to actual execution. We represent devices, tasks, and operating rules as models and execute them according to defined semantics.

As large language models become involved in laboratory decision-making and control, we also need to define how their requests enter device execution.

Slide 2 Presentation Overview

The presentation has six parts. I will start with the background and the execution boundary problem. Then, I will introduce the unified modeling framework and show how these models work together.

Next, I will use a reactor heating task to show the complete execution sequence. Finally, I will discuss the simulation feedback loop and summarize our work.

Let’s first look at the background.

Slide 3 Background and Execution Boundaries

As large language models advance, smart chemical laboratories are gradually evolving from automated workflow execution toward large language model involvement in decision-making and control.

But whether an execution intent comes from a person, a workflow, or a large language model, we need to explain four things:

How are resources organized and accessed?

How does device state evolve?

How is data generated and transferred?

Does execution stay within allowed limits?

This issue exists even without large language models, but their involvement makes execution boundaries more pressing.

So, we need to carefully consider what kinds of control requests large language models can make and under what conditions these requests can be executed.

Our principle is simple: large language models must not have a privileged bypass. This means that all control requests follow the same execution path.

Therefore, we need an executable framework that keeps resource organization, state evolution, data transfer, and boundary checks aligned.

The next question is how to organize these models.